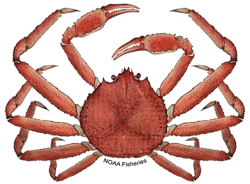
# *Environmental stress and hybridization in Bering Sea crabs* (Chionoecetes *spp.*)



**Annual Shellfish Meeting, NSA-PCS & PCSGA**
**September 2026, Blaine WA**

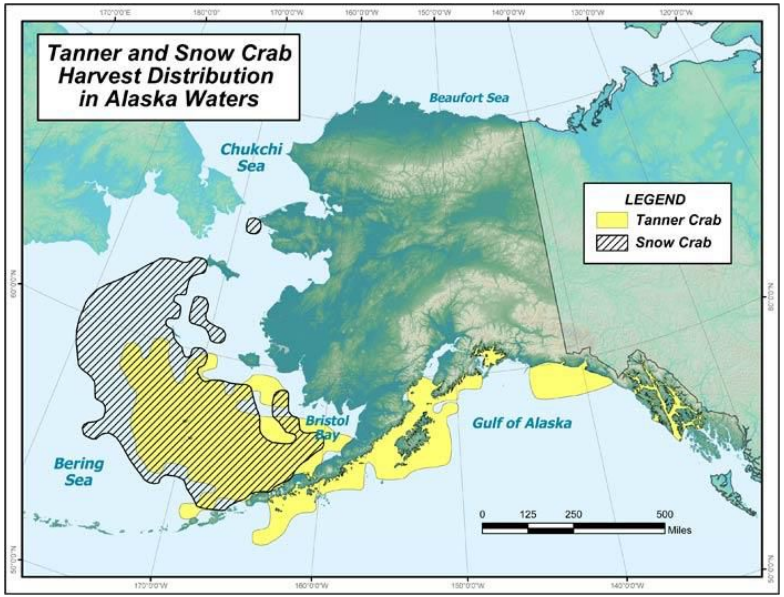

*Laura Spencer, NOAA Affiliate at Univ. of Washington*
*Jennifer Gardner, NOAA AFSC Kodiak*
*Ingrid Spies, NOAA AFSC Seattle*
*Steven Roberts, University of Washington SAFS*
*Chris Long, NOAA AFSC Kodiak*

# Tanner & snow crab (*Chionoecetes spp.*) in the Bering Sea







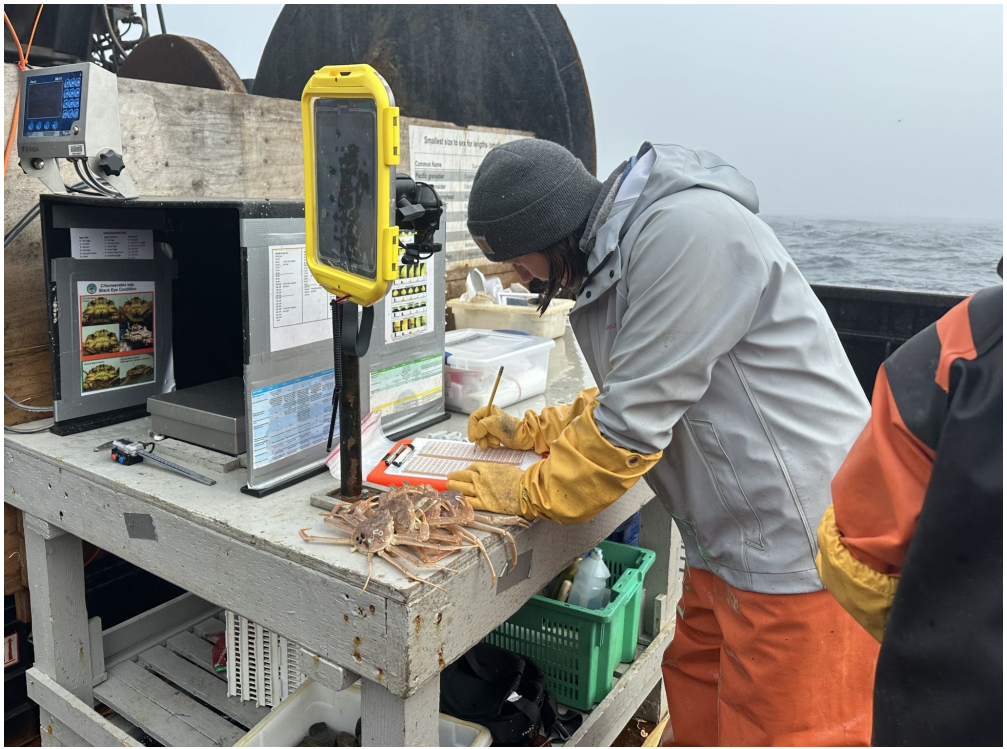



*Credit: ADFG*

*They hybridize, F1's are viable*

# *Crab stocks monitored with annual*
# *Eastern Bering Sea Bottom Trawl Survey*

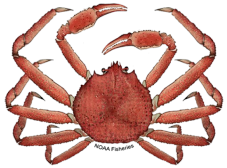




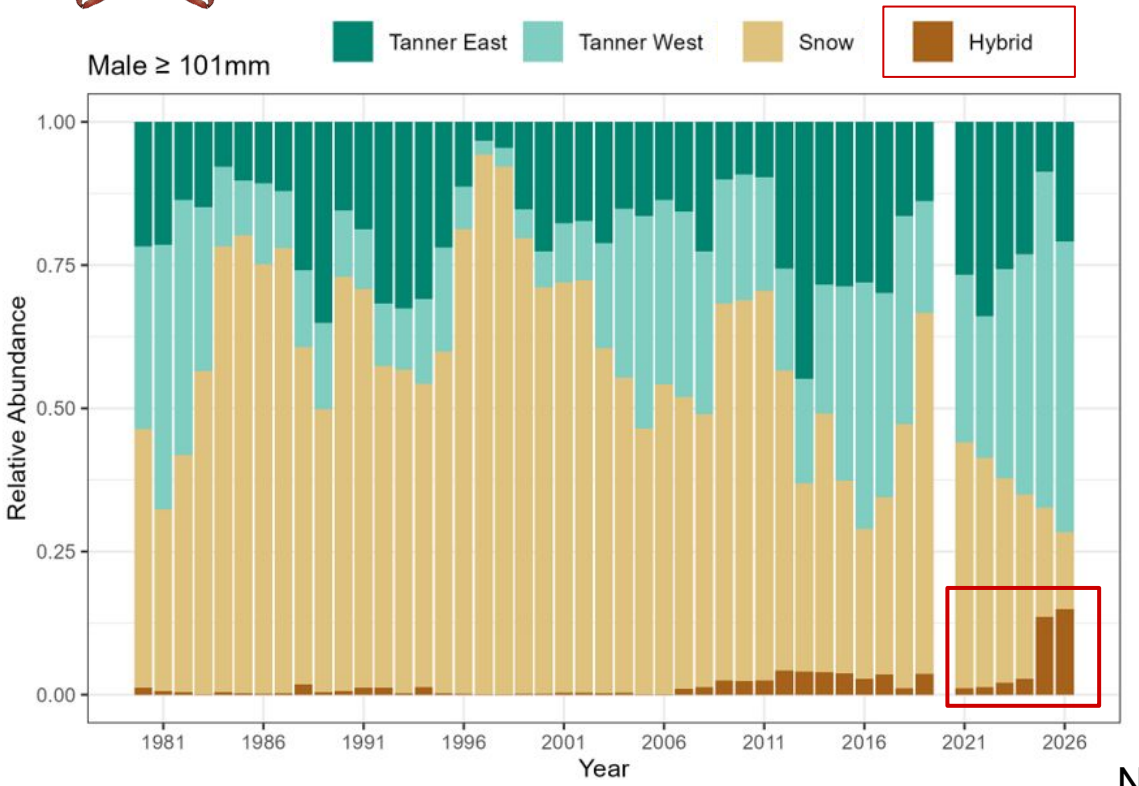
*Stratified random sampling design*

*June-August at 350 stations, ½-hr tows*
*Species, sex, catch weight, size composition, shell condition, maturity, reproductive status*

*Images: AFSC*



# Snow-Tanner hybrids are becoming more abundant




*Hybrids are identified visually on survey using 4 traits, e.g. eye color*

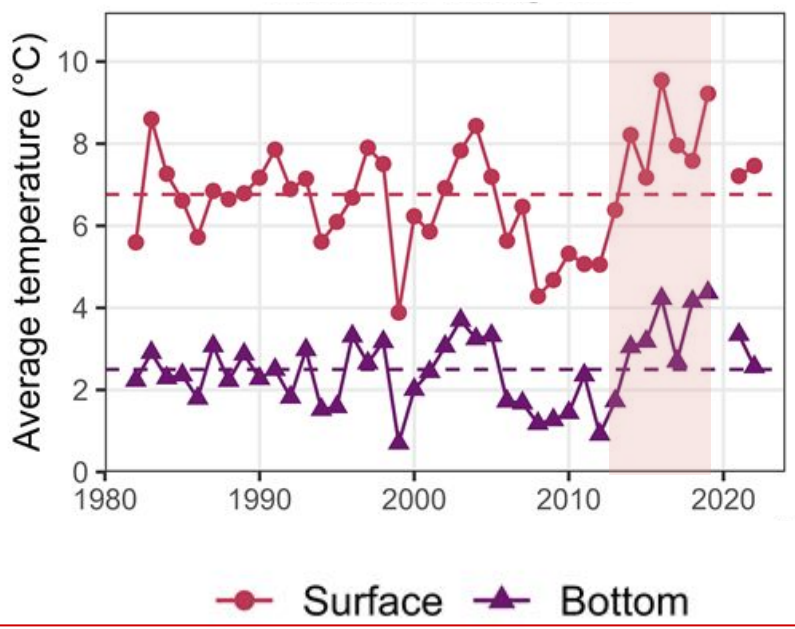




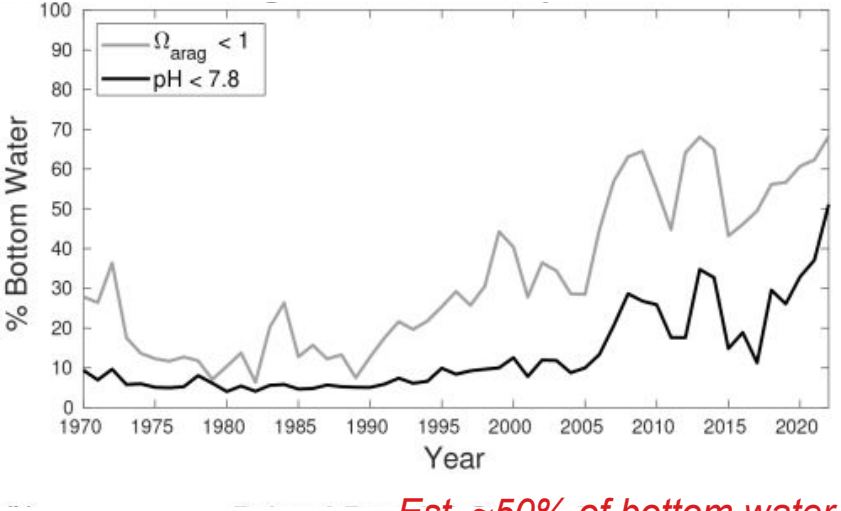

No hybrid fishery! Management issue.

# **Environmental change** is likely impacting crab populations in the Bering Sea

*Warming!*

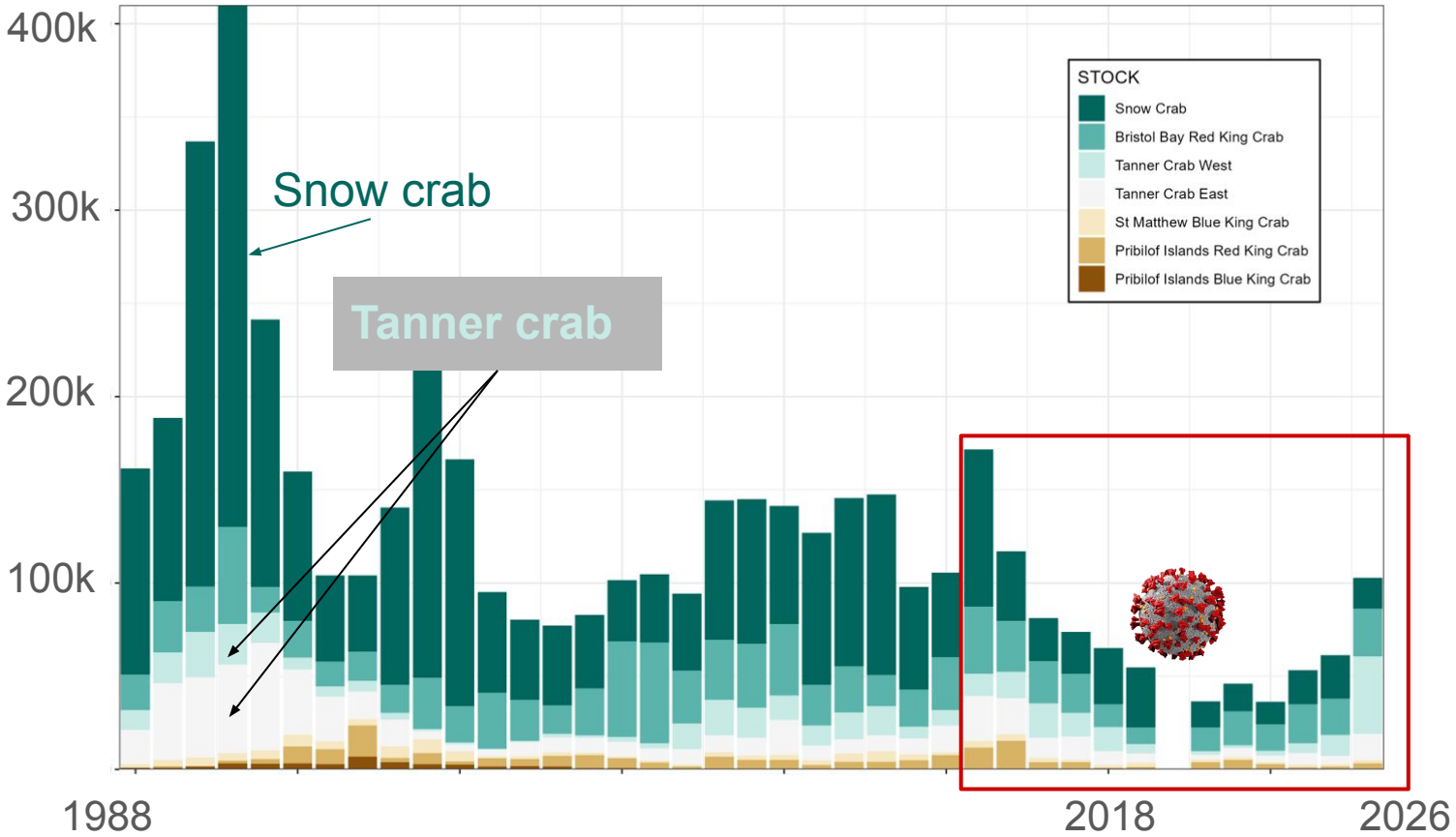



*Acidification!*



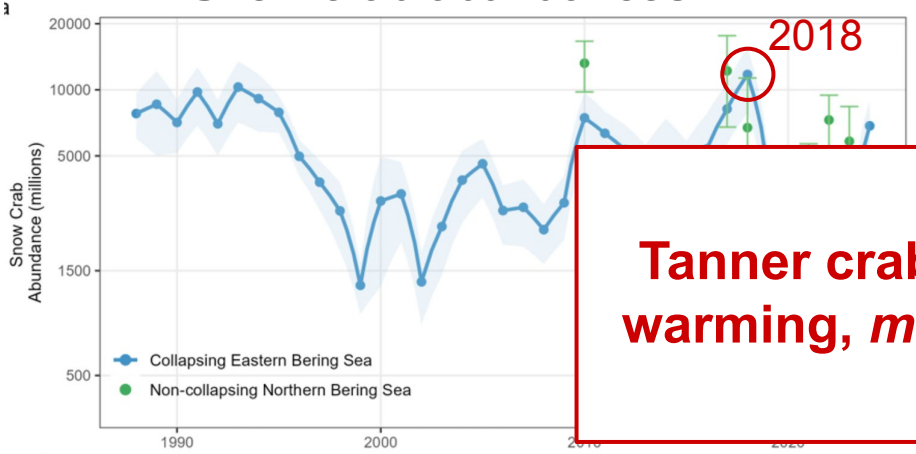

*Est. ~50% of bottom water pH < 7.8 in summer*

# Harvestable crab biomass (t) have decreased in the past decade

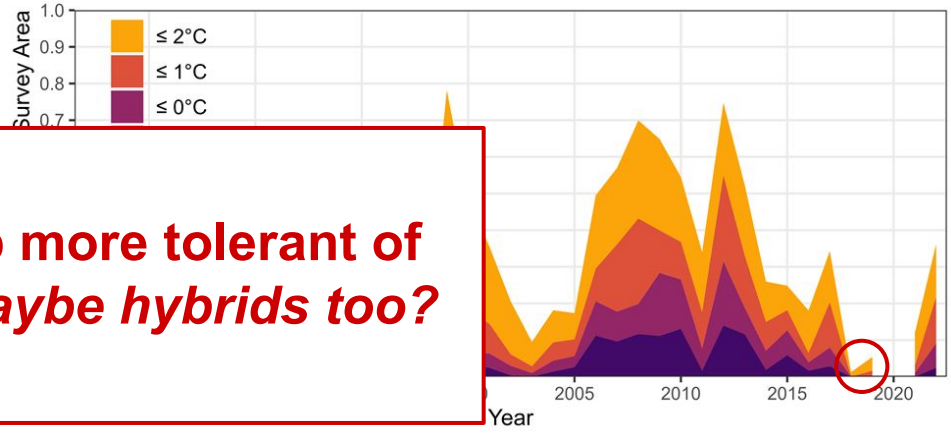


# Est. 10 billion snow crab died between 2018 – 2021, fisheries closed in 2022, 2023

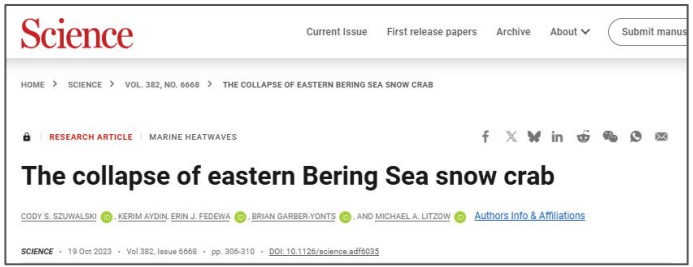
*Snow crab abundances*




*EBS cold pool extent*




**Tanner crab more tolerant of warming, *maybe hybrids too?***


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RESEARCH ARTICLE | MARINE HEATWAVES

**The collapse of eastern Bering Sea snow crab**

CODY S. SZUWALSKI, KERIM AYDIN, ERIN J. FEDEWA, BRIAN GARBER-YONTS, AND MICHAEL A. LITZOW Authors Info & Affiliations

SCIENCE · 19 Oct 2023 · Vol 382, Issue 6668 · pp. 306-310 · DOI: 10.1126/science.adf6035

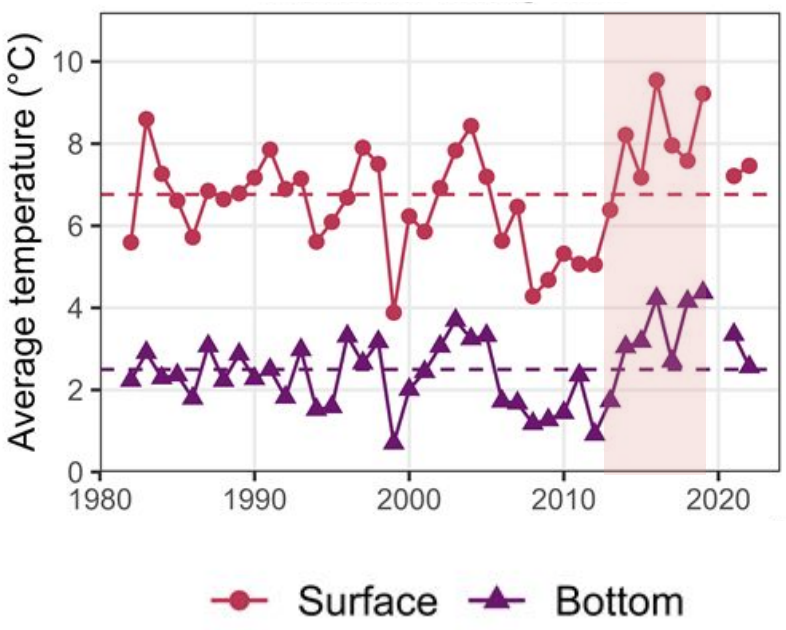
**Energetic limitations and mass mortality of Bering Sea snow crab: Interacting effects of warming and density on collapse and recovery**

Authors: Erin J. Fedewa, Louise A. Copeman, and Michael A. Litzow | AUTHORS INFO & AFFILIATIONS

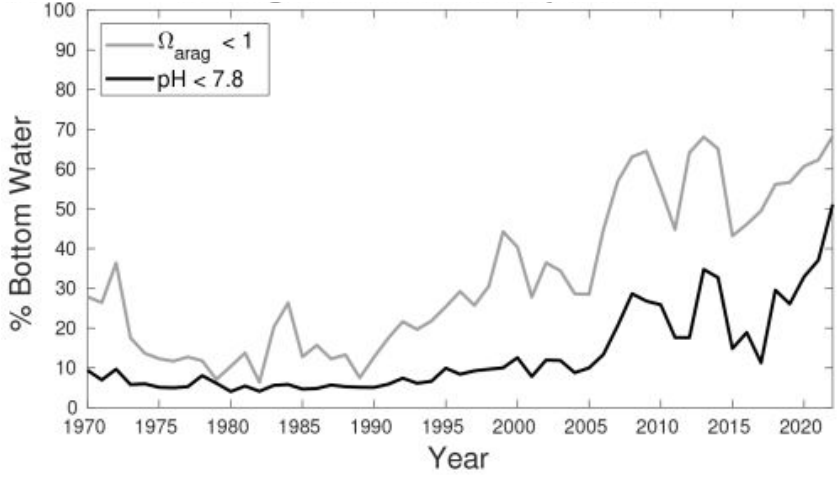
Publication: Canadian Journal of Fisheries and Aquatic Sciences • 24 November 2025 • https://doi.org/10.1139/cjfas-2025-0099

# **Environmental change** is likely impacting crab populations in the Bering Sea

*Warming!*



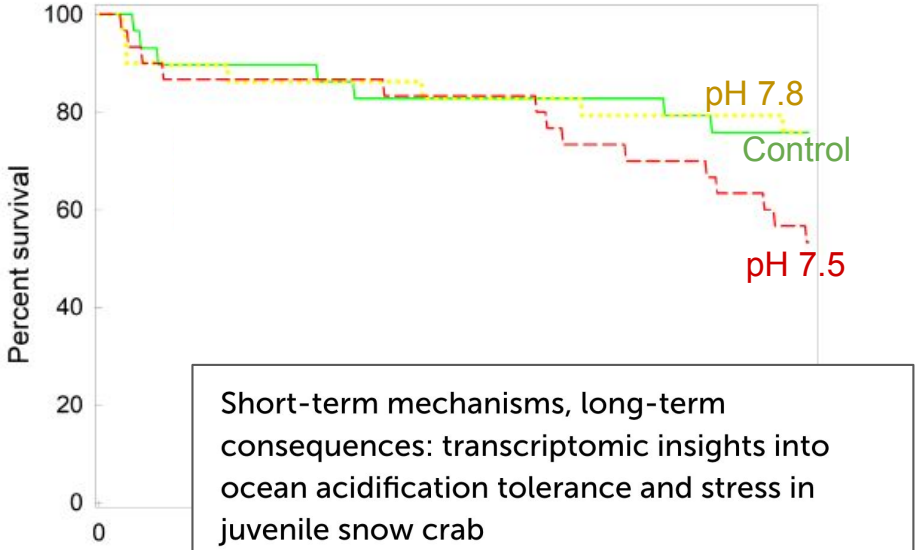

*Acidification!*




*Est. ~50% of bottom water pH < 7.8 in summer*

# Snow crab are less sensitive to Ocean Acidification than Tanner crab, *what about hybrids*?

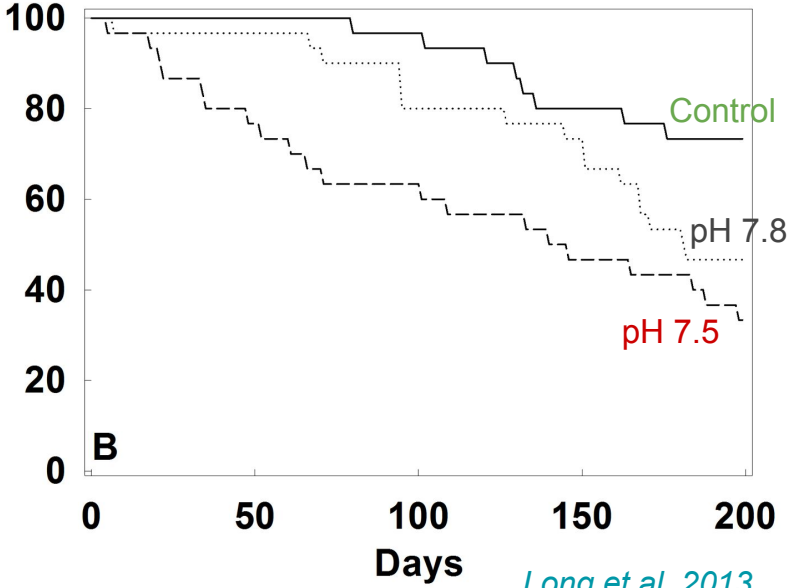
## *Snow crab*




Short-term mechanisms, long-term consequences: transcriptomic insights into ocean acidification tolerance and stress in juvenile snow crab

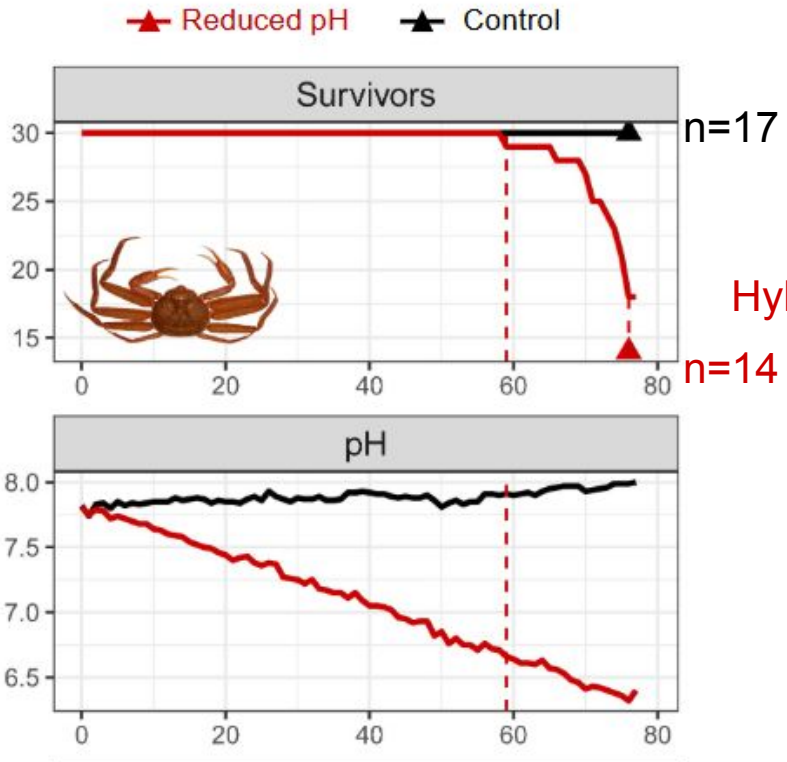
Laura H. Spencer [1,2*] Ingrid Spies [2] Jennifer L. Gardner [3‡]
Steven B. Roberts [1] W. Christopher Long [3]

## *Tanner crab*




Long et al. 2013

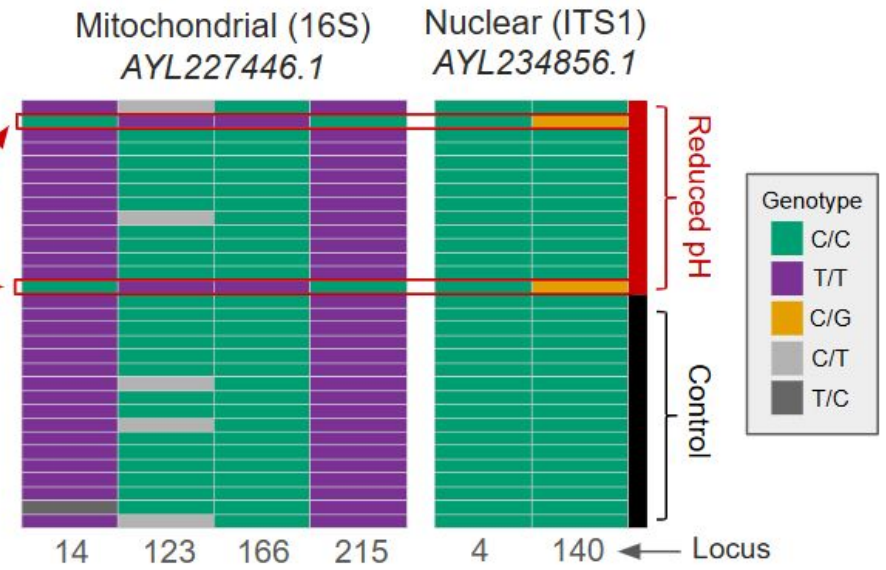
# Unexpected guests – Hybrids snuck into Tanner severe low pH selection study and survived



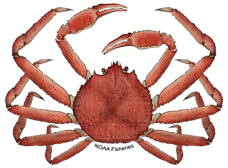

n=17

n=14

Hybrids




***Maybe hybrids are more OA tolerant?***



# Hybrids may play a larger role as the Bering Sea continues to change, but many questions



**Why so many hybrids?**

- Do hybrids have higher climate stress tolerance?
- Is there a redistribution in mating ratios?
- *NOAA modeling study*: how does the marine heatwave and snow crab collapse explain hybrid prevalence?

**Can hybridization be a source of adaptive genetic variation?**

- *Planned study*: Develop genetic tool to identify hybrids (F1, F2, backcrosses), map their distribution; environmental association?

# Bottom Trawl Survey & Experimental Teams are *testing visual hybrid ID consistency, accuracy*



***Trial happening now***

- ~40 crabs brought back from survey & tagged
- All staff assign crabs to snow, Tanner, hybrid
- Will confirm with genetic ID

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